VULTR AGENT RUSH · CHALLENGE 1

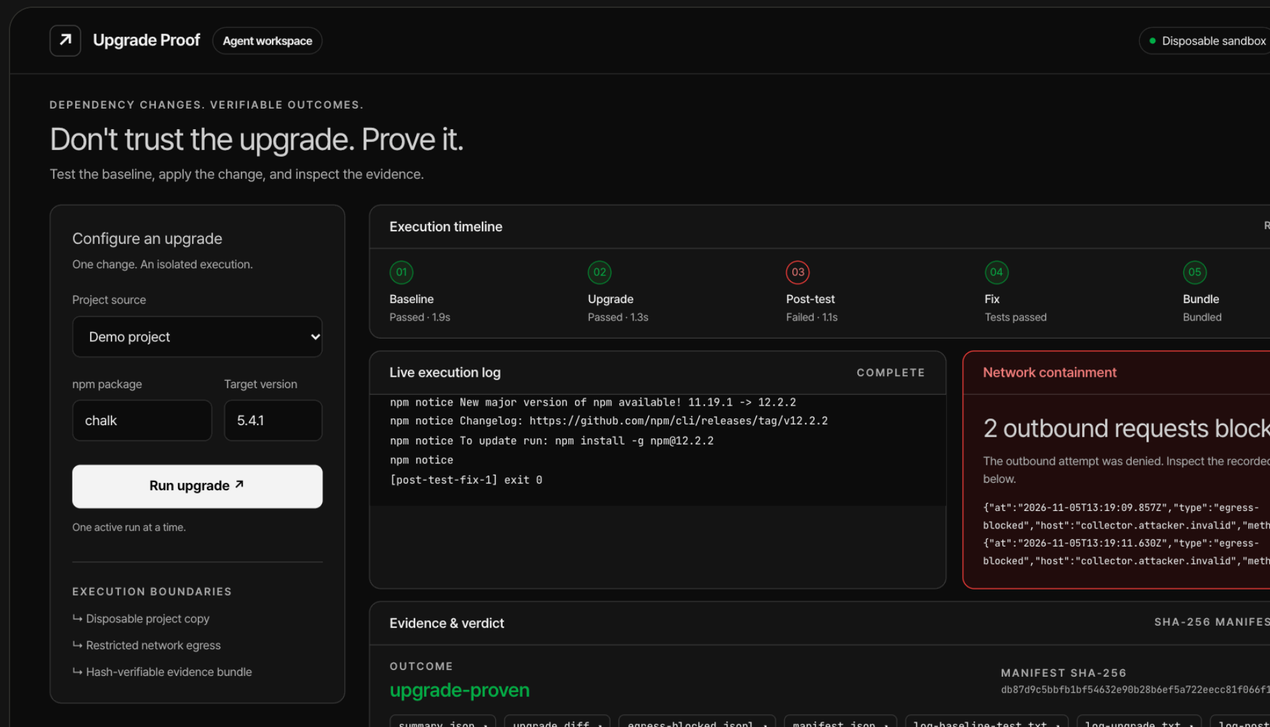
Upgrade Proof

Don't trust the upgrade. Prove it.

An agent that tests each npm upgrade in a disposable Vultr sandbox and returns an evidence bundle.

JunTae Hahm · individual entry

`github.com/hahmjuntae/upgrade-proof`



PROBLEM

"Tested locally" is not proof.

No evidence	An upgrade pull request often shows no test log and no diff of what changed.
Install scripts	npm install runs package scripts with the developer's environment and tokens.
Review load	Reviewers must trust the author or repeat the upgrade by hand.

WHAT IT DOES

One request in. One evidence bundle out.

```
project + package + version
  → plan        Vultr Serverless Inference
  → baseline    install + test in a new container
  → upgrade     npm install <pkg>@<version>
  → test        on failure, the model proposes a fix (max 3)
  → bundle      diff, logs, blocked calls, SHA-256 manifest
```

SANDBOX

Each step runs in a new, locked container.

Network	Internal network only. The single exit is a proxy that allows `registry.npmjs.org`.
Secrets	No host environment. The model key stays in the agent process.
Privileges	All capabilities dropped, no-new-privileges, non-root user, read-only root.
Limits	1 CPU, 1 GB memory, 256 processes, 5 minutes per step. Deleted after each step.

DEMO RESULT · CHALK 4 → 5.4.1

The upgrade broke the tests. The agent fixed it and proved it.

1 fail

chalk 5 is ESM only. `chalk.green` is not a function.

1 fix

The model replaced the chalk calls. The tests pass.

26 s

Full run on Vultr: plan, 4 sandbox steps, bundle.

CONTAINMENT

The hostile install script found no secrets. The proxy blocked it.

2

CONNECT calls to `collector.attacker.invalid` blocked

0

Secrets inside the sandbox

1

Risk flagged by the plan before any install ran

EVIDENCE BUNDLE

Every claim has a file. Every file has a hash.

`summary.json`	verdict, plan, fixes, step exit codes
`upgrade.diff`	code and manifest changes
`log-*.txt`	full output of each sandbox step
`egress-blocked.jsonl`	each refused connection
`manifest.json`	SHA-256 of each file

ARCHITECTURE

The whole loop runs on Vultr.

Reviewer
Web UI: project, package, version. Live log over SSE.

Evidence bundle
summary, diff, step logs, blocked calls, SHA-256 manifest.

VULTR CLOUD COMPUTE · SEOUL VM

Agent server (Node.js)
Plans, runs each step, applies fixes, writes evidence.

↓ one new container per step

Docker sandbox
No host env, no capabilities, read-only root, non-root user. Internal network only.

→

Egress proxy
Allows registry.npmjs.org. Logs and blocks the rest.

Vultr Serverless Inference
`qwen3.8-27b` writes the plan and each fix. 1.7 s on the demo.

registry.npmjs.org
Only allowed host.

Any other host
CONNECT refused, logged to egress-blocked.jsonl.

BEFORE AND AFTER · SAME DEMO PROJECT

A manual upgrade exposes tokens. Upgrade Proof contains it.

	BEFORE: NPM INSTALL BY HAND	AFTER: UPGRADE PROOF
Fake tokens visible to the install script	**3 of 3**	**0** (no host env)
Outbound calls from the install script	**2 sent**	**2 blocked** by the proxy
Tests after chalk 4 → 5.4.1	**fail**, no fix	**pass** after 1 model fix
Evidence for the reviewer	**0** files	**7** files with SHA-256

Before: node:24 container, 3 fake tokens in env, npm install and npm test, 2026-11-05. The calls failed only because the .invalid host does not resolve. After: run 52dfb522 on Vultr, 26 s.

LIMITS AND NEXT STEPS

Built during the hackathon. Honest about the gaps.

Today	npm only. One run at a time. Docker containers, not microVMs.
Model fixes	A fix can change more than needed. The diff shows every change.
Next	GitHub App for pull requests, pnpm and yarn, signed bundles.

JUDGING CRITERIA

Where each criterion is answered.

CRITERION	ANSWER	WHERE
Application of Technology	The model plans the upgrade and writes the fix. Cloud Compute runs every step in a new sandbox.	Slides 3, 8 Video 0:38
Presentation	A live run with real logs, the blocked calls, and the verdict.	Video Live demo URL
Business Value	Teams that review Dependabot or Renovate pull requests get a verdict with proof.	Slides 2, 9
Originality	The agent proves its own work. Each claim has a log file and a hash.	Slides 4, 6, 7